



WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**

WPS No. **RNC-WPS-017**

Date **12-Apr-13**

Revision No. **0**

Supporting PQR No.(s) **RNC-PQR-017**

Date **12-Apr-13**

Type(s) **Manual**

Welding Process(es) **Gas Tungsten Arc Welding + Shield Metal Arc welding**

(Automatic, Manual, Machine, or Semi-Auto.)

(GTAW+SMAW)

JOINT (QW-402)

Joint Design **Single V**

Backing (Yes) ☐ (No) ☒ X

Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal

☐ Nonmetallic ☐ Other

Notes

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **1** Group No. **1 or 2**

to P-No. **1**

Group No. **1 or 2**

Material Specification Type and Grade

A106 Gr.B, A234 WPBEquivalents

TO **A106 Gr.B, A234 WPBEquivalents**

Chem. Analysis and Mech. Prop.

Thickness Range :

Base Metal :

Groove

5 mm -42.88 mm.

Fillet

All

Pipe Dia. Range :

Groove

NPS 2 7/8"-Unlimited

Fillet

All

Other

FILLER METALS (QW-404)

Spec. No. (SFA)

GTAW

5.18

SMAW

5.1

AWS No. (Class)

ER 70S-G

E 7016

F-No.

6

4

A-No.

1

1

Size of Filler Metals

Ø 2.4 mm

Ø 3.2 & 4.0 mm.

Deposited Weld Metals

5.0 mm.

16.44 mm.

Thickness Range :

Groove

10.0 mm Max.

32.88 mm. Max.

Fillet

ALL

ALL

Electrode-Flux (Class)

-

-

Flux Trade Name

-

-

Consumable Insert

-

-

Other

Kobelco TGS-50 or Equivalents

Kobelco LB 52 or Equivalents

POSITION (QW-405)

Position(s) of Groove

ALL

Welding Progression

Up

X

Down

Position(s) of Fillet

ALL

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range

595°-625°C

Time Range

Holding Time 1 Hour

Notes

Cooling Rate = 250° Max.

PREHEAT (QW-406)

Preheat Temp.

Min.

T ≤ 25 mm. = 10°C

Interpass Temp.

Max.

T ≥ 25 mm. = 79°C

250°C

Preheat Maintenance

N/A

Notes

GAS (QW-408)

Percent Composition

Shielding

Trailing

Backing

Gas(es)	(Mixture)	Flow Rate
Argon	99.99%	12-16 Ltr/Min
-	-	-
-	-	-

WPS No. RNC-WPS-017 Rev. 0

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☒ AC ☒ DC Polarity EN(GTAW)+EP(SMAW)
 Amps (Range) 100-140(GTAW) + 90-120(SMAW) Volts (Range) 10-13(GTAW)+ 27-37(SMAW)
 Tungsten Electrode Size and Type Ø 2.4 mm , Thoriated (EWTh-2)
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS
 Orifice or Gas Cup Size 8.0 mm - 19.0 mm
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
 Method of Back Gouging N/A
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE
 Travel Speed (Range) 8-12 (GTAW)+5-10(SMAW)Cm/Minute
 Peening N/A
 Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER 70S-G	2.4	DC EN	100-130	10-11	8-10	
2nd	GTAW	ER 70S-G	2.4	DC EN	120-140	10-13	10-12	
3 & Over	SMAW	E 7016	3.2 & 4.0	DC EP/AC	90-120	27-37	5-10	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of ASME-IX, 2007 latest Revision.

Prepared by: Mr. Chatchawan S. QA/QC RNC 30/04/13
 Name Signed Date
 Reviewed by: ภูชิน สีหมอนไชย 30/04/13
 Name Signed Date
 Approved by: ภูชิน สีหมอนไชย 30/04/13
 Name Signed Date

Procedure Qualification Record (PQR)

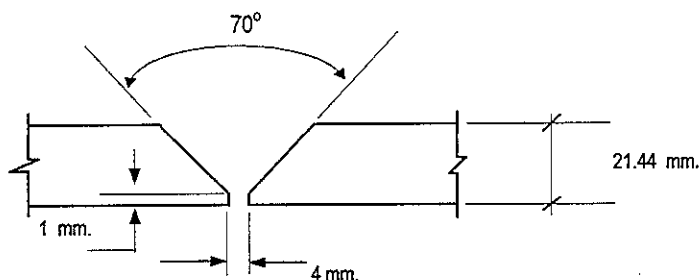
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : RNC-PQR-017

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-017	Date :	12 April 2013
WPS Number :	RNC-WPS-017		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding (SMAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)							
Material Spec.		SA 106	To	SA 106		Temperature		610° C			
Type or Grade		Gr.B	To	Gr.B		Hold Time		1 HR.			
P. No.		1	To	1		Others		Cooling Rate 135 °C/Hr.& Heating Rate 180° C/Hr.			
Thickness of Test Coupon				21.44 mm.							
Diameter of Test Coupon				Ø 10"							
Others				GAS (QW-408)							
				Percent Composition							
					Gas (es)		Mixture		Flow Rate		
				Shielding	Argon		99.95%		14 L/Min.		
				Trailing	-		-		-		
				Backing	-		-		-		
FILLER METALS (QW-404)				GTAW		SMAW		ELECTRICAL CHARACTERISTICS (QW-409)			
SFA Specification		A 5.18		A 5.1		Current		DC (GTAW)		DC (SMAW)	
AWS Classification		ER 70S-G		E 7016		Polarity		EN		EP	
Filler Metal F-No.		6		4		Amperage		100 - 125 (GTAW)		Voltage 10 - 11 (GTAW)	
Weld Metal Analysis A-No.		1		1				118 - 125 (SMAW)		24 - 25 (SMAW)	
Size of Filler Metal		Ø 2.4 mm.		Ø 3.2 mm.		Tungsten Electrode Size		Ø 2.4 mm. 2% thoriated.EWTh-2			
Others		Kobelco TGS-50		Kobelco LB 52		Others		N/A			
Weld Metal Thickness		-									
POSITION (QW-405)						TECHNIQUE (QW-410)					
Position of Groove		6G				Travel Speed		5.68 - 11.38 cm/min.			
Progression of Welding (Uphill or Down)		Up Hill				String or Weave Bead		Both			
Others		N/A				Oscillation		N/A			
PREHEAT (QW-406)						Multiple or single Pass		Multiple Pass			
Preheat Temperature		36° C				Multiple or Single Electrodes		Multiple Electrodes			
Interpass Temperature (max.)		180° C				Others		N/A			
Others		N/A									

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- RNC-PQR-017

Tensile Test Report No.TS-13-04-092

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
1184/13 TR1	19.46	20.33	395.622	202.213	511.127	OUT OF WELD
1184/13 TR2	19.35	20.10	388.935	202.017	519.411	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-04-042

Type and Figure Number	Type of Bend	Indication	Results
1184/13 BS1	Transverse face bend	No Open Discontinuity	Accepted
1184/13 BS2	Transverse face bend	No Open Discontinuity	Accepted
1184/13 BS3	Transverse root bend	No Open Discontinuity	Accepted
1184/13 BS4	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

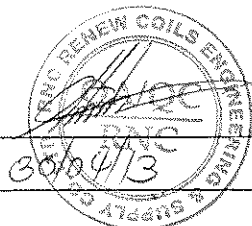
Type of Test Radiographic examination results Accepted Report No.PQR-007
Hardness Testing Report No. HV-13-04-023
Macrostructure Report No. MA-13-04-108
Impact Testing Report No.IM-13-04-031 (-40°C)

Welders Name Mr.Narongkorn Y Welder Number W-013
Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date :



Inspected by : Mr.Wichit K

Certified by : Mr.Chaloemkiet R.

Date : 27 April 2013





PAE TECHNICAL SERVICE CO.,LTD

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 682208-9

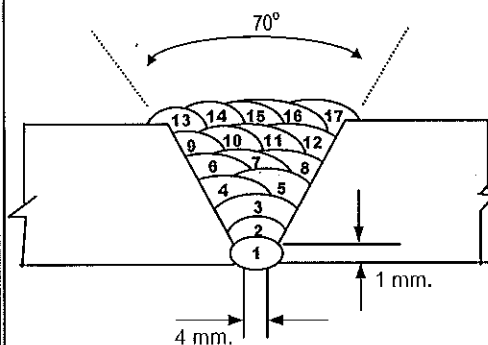
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 2

Client :	Project Name : MINI SHUTDOWN PTTGC	Location : RNC Work Shop	Report No. PQR-004
Renew Coils Engineering & Supply Co.,Ltd.	Customer Order No. : N/A	WPS No. : RNC-WPS-017	Date : 10-Apr-13

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
W-013	MR.NARONGKORN Y.	GTAW	6G	ASME IX	A106 Gr.B	ø 10" Thk. 21.44 mm.	A5.18	ACCEPTED	ACCEPTED
		SMAW					A5.1		

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER 70S-G	2.4	DCEN	100	10	5.68	36	10.57	Argon	14	-	-
2	GTAW	ER 70S-G	2.4	DCEN	125	11	9.70	120	8.51	Argon	14	-	-
3	SMAW	E 7016	3.2	DCEP	119	24	7.96	135	21.53	-	-	-	-
4	SMAW	E 7016	3.2	DCEP	120	24	8.51	164	22.31	-	-	-	-
5	SMAW	E 7016	3.2	DCEP	122	24	8.61	180	20.41	-	-	-	-
6	SMAW	E 7016	3.2	DCEP	121	24	9.48	157	18.38	-	-	-	-
7	SMAW	E 7016	3.2	DCEP	122	24	9.62	163	18.27	-	-	-	-
8	SMAW	E 7016	3.2	DCEP	124	24	9.75	154	18.32	-	-	-	-
9	SMAW	E 7016	3.2	DCEP	125	25	9.82	140	19.10	-	-	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :

Mr. Wichit K.

11-Apr-2013

SIGNATURE :

NAME :

DATE :

Mr. Chatchawan S.

30/04/13

SIGNATURE :

NAME :

DATE :

Phuchin Seehomchai

Phuchin Seehomchai



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NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 682208-9

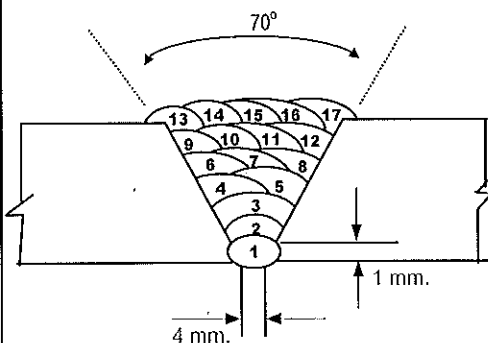
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 2 of 2

Client :	Project Name : MINI SHUTDOWN PTTGC	Location : RNC Work Shop	Report No. PQR-004
Renew Coils Engineering & Supply Co.,Ltd.	Customer Order No. : N/A	WPS No. : RNC-WPS-017	Date : 10-Apr-13

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
W-013	MR.NARONGKORN Y.	GTAW	6G	ASME IX	A106 Gr.B	ø 10" Thk. 21.44 mm.	A5.18	ACCEPTED	ACCEPTED
		SMAW					A5.1		

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
10	SMAW	E 7016	3.2	DCEP	120	24	8.85	149	19.53	-	-	-	-
11	SMAW	E 7016	3.2	DCEP	118	24	8.58	138	19.81	-	-	-	-
12	SMAW	E 7016	3.2	DCEP	121	24	8.69	152	20.05	-	-	-	-
13	SMAW	E 7016	3.2	DCEP	120	24	9.48	145	18.23	-	-	-	-
14	SMAW	E 7016	3.2	DCEP	123	24	9.64	141	18.38	-	-	-	-
15	SMAW	E 7016	3.2	DCEP	122	24	9.36	132	18.77	-	-	-	-
16	SMAW	E 7016	3.2	DCEP	120	24	11.38	161	15.19	-	-	-	-
17	SMAW	E 7016	3.2	DCEP	123	24	9.60	152	18.45	-	-	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :

Mr. Wichit K.

11-Apr-2013

SIGNATURE :

NAME :

DATE :

Mr. Chatchawan S.

30/04/13

SIGNATURE :

NAME :

DATE :

Phuchin Seehomchai

Phuchin Seehomchai

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)682208-9, Fax : 038)682206, SRIRACHA : 038)330220, 330229

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PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)682208-9, Fax : 038)682206, SRIRACHA : 038)330220, 330229

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnmp.co.th



0301

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-04-092

Revision No. 0

Test Product	PQR No. RNC-PQR-017	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Test	Reduced Section Tension
Dimension	Ø 10" #120 x 21.44 mm (Thk.)	Equipment's Capacity	200 TONS
Date Received	23 April 2013	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	25 April 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	1184/13 TR1	N/A	20.33	19.46	395.622	N/A	N/A	N/A	202.213	511.127	N/A	OUT OF WELD
2	1184/13 TR2	N/A	20.10	19.35	388.935	N/A	N/A	N/A	202.017	519.411	N/A	OUT OF WELD

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-017

Welding Process : GTAW+SMAW

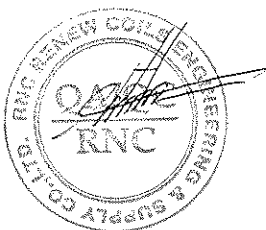
Filler/Electrode Spec't : ER70S-G E7016

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date / /

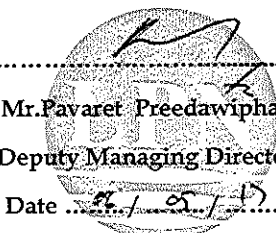


Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date / /



LPN PLATE MILL

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Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpr.co.th



0302

TESTING
No. 01/98

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-04-042

Revision No. 0

Test Product	PQR No. RNC-PQR-017	Material Characterization	Pipe
Project Name	N/A	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 10" #120 x 21.44 mm (Thk.)	Diameter of Meudrel	40 mm
Date Received	23 April 2013	Bend Angle	180 Degree.
Date Tested	25 April 2013	Equipment/ Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	1184/13 BS1	Transverse side bend	10.00x19.96x160.32 mm	No Open Discontinuity	N/A	-
2	1184/13 BS2	Transverse side bend	9.98x19.89x160.89 mm	No Open Discontinuity	N/A	-
3	1184/13 BS3	Transverse side bend	9.94x19.99x161.04 mm	No Open Discontinuity	N/A	-
4	1184/13 BS4	Transverse side bend	9.93x19.90x160.21 mm	No Open Discontinuity	N/A	-

Additional details : WPS No. RNC-WPS-017

Welding Process : GTAW+SMAW

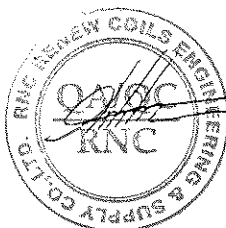
Filler/Electrode Spec't : ER70S-G E7016

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date / /



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date / /

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Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnrm.co.th



Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-04-031

Revision No. 0

Test Product	PQR No. RNC-PQR-017	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a ¹
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Test	Charpy V-Notch Impact
Dimension	Ø 10" #120 x 21.44 mm (Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	23 April 2013	Equipment 's Capacity	542 Joules
Date Tested	25 April 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Specimen Dimension			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	1184/13-1	-40°C	10.012	10.004	54.80	7.05	9.87	Base Metal	-
2	1184/13-2	-40°C	10.008	10.005	54.81	14.58			
3	1184/13-3	-40°C	10.000	10.004	54.85	7.99			
4	1184/13-1	-40°C	10.009	10.008	54.90	48.85	68.33	HAZ	-
5	1184/13-2	-40°C	10.009	10.006	54.90	82.25			
6	1184/13-3	-40°C	10.011	10.005	54.86	73.89			
7	1184/13-1	-40°C	10.006	10.007	54.86	63.35	38.58	Weld Metal	-
8	1184/13-2	-40°C	10.012	10.008	54.90	26.90			
9	1184/13-3	-40°C	10.008	10.008	54.94	25.50			

Additional details : WPS No. RNC-WPS-017

Welding Process : GTAW+SMAW

Filler/Electrode Spec't : ER70S-G E7016

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date/...../.....

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date/...../.....

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LPN PLATE MILL



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

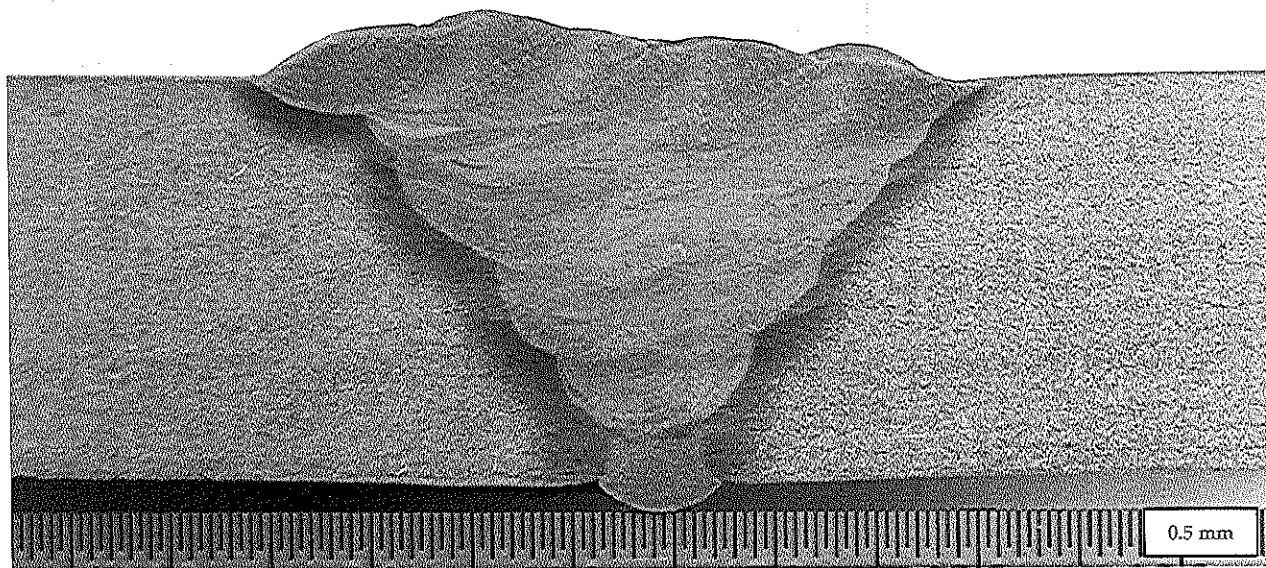
Macrostructure Analysis Report**Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.****Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.**

Page 1 of 1

Report No. MA-13-04-108

Revision No.0

Test Product	PQR No.RNC-WPS-017	Material Characterization	Pipe to Pipe
Test No.	1184/13 MA	Equipment used	Microscope upto 40X (times)
Test Location	-	Serial No.	LEICA 0650854912
Material Specification	A106 Gr.B	Method of Preparation	ASME Section IX : 2010 edition
Dimension	Ø10"#120 THK21.44	Test Method	ASME Section IX : 2010 edition
Welding Process/Position	GTAW+SMAW	Standard of Test	ASME Section IX : 2010 edition
Type of Joint	Butt Joint	Etchant Reagent	25 ml HNO ₃ + 75 ml H ₂ O
Welder Name/ No.	N/A	Time of Etching	5 minutes
Date Received	23 April 2013	Date Tested	25 April 2013



Macro-etch Examination : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

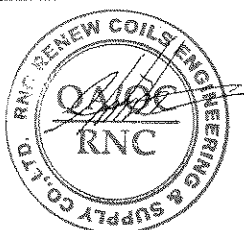
Additional details : -

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date : 27/05/13

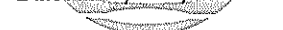


Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date : 27/05/13

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnrm.co.th

13314**Hardness Testing Report**

Page 1 of 1

Report No. HV-13-04-023

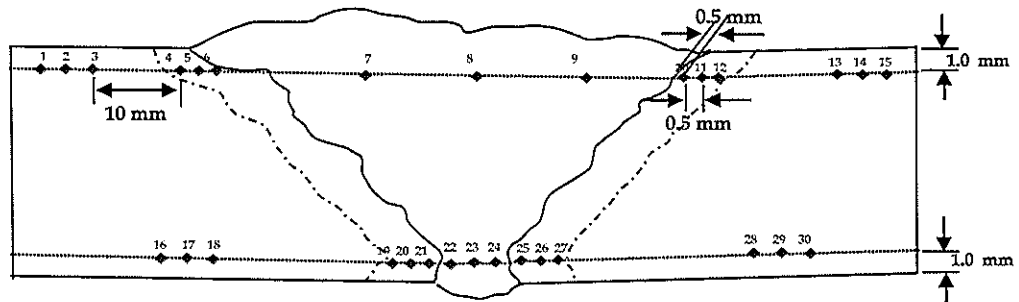
Revision No. 0

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.**Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000 Thailand.**

Test Product	PQR No.RNC-WPS-017	Material Characterization	Pipe to Pipe
Test No.	1184/13 HN	Test Method	ASTM E92-82 (Reapproved 2003) ^{E2}
Project Name	N/A	Standard of	ASME section IX: 2010 edition
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A106 Gr.B	Equipment	Wilson Wolpert , 432 SVD (V32D219)
Dimension	Ø10" #120 THK21.44	Indenter Size	Pyramid diamond, angle of 136°
Welding Process/Position	GTAW+SMAW	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	23 April 2013	Cal. Block	No. A60023 No. A62072
Date Tested	25 April 2013	Ambient Temperature	26.9 °C

Test Location

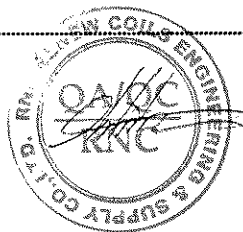
Total = 30 points



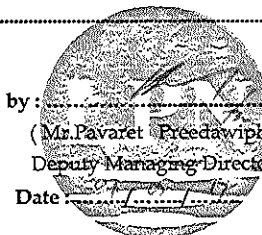
Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale)
1	BASE	143.8	16	BASE	147.6	*****		
2	BASE	144.1	17	BASE	148.8			
3	BASE	144.3	18	BASE	148.9			
4	HAZ	154.5	19	HAZ	167.6			
5	HAZ	181.0	20	HAZ	170.9			
6	HAZ	206.7	21	HAZ	170.0			
7	WELD	169.2	22	WELD	154.1			
8	WELD	164.6	23	WELD	156.7			
9	WELD	153.8	24	WELD	159.0			
10	HAZ	165.9	25	HAZ	173.0			
11	HAZ	152.6	26	HAZ	172.2			
12	HAZ	148.4	27	HAZ	174.3			
13	BASE	147.7	28	BASE	146.0			
14	BASE	148.0	29	BASE	147.5			
15	BASE	146.5	30	BASE	148.1			

Additional details : -

Reported by :
 (Mr.Keerati Sa-ngoen)
 (Acting) General Manager - Quality Assurance
 Date : 27/4/13



Approved by :
 (Mr.Pavaret Freedawiphat)
 Deputy Managing Director
 Date : 27/4/13

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LPN PLATE MILL

004

NUMBER 45



Page : 1

Report No. : PW-003

Project No. :

Location :	☒ -Shop	☐ -Field	Recorder No.:	EH-07ZA194
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Method Type: ☒ -Local ☐ -Furnace ☐ -Internal Firing

Completion Date: 15-04-2013

Procedure No.

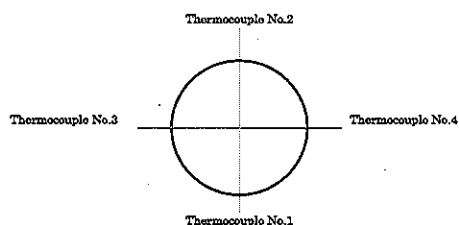
HEAT TREATMENT CONDITION/RESULTS

Description	Loading Temp[1] (°C)	Heating Rate[2] (°C/Hr.)	Holding Temp[3] (Min.-Max.)	Holding Time[4] (Hr.)	Cooling Rate[5] (°C/Hr.)	Unloading Temp[6] (°C)
Request	300	200	595-625	1 HR.	250	300
Actual	300	180	610	1 HR.	135	300
Request						
Actual						
Request						
Actual						

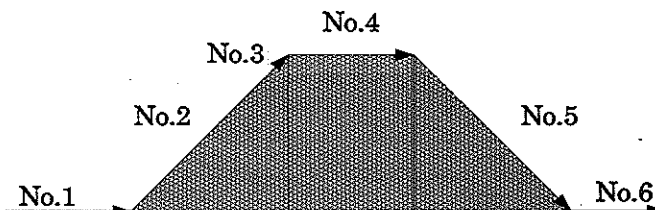
DESCRIPTION

Line No./Dwg.No./Class No./Sht.no./Spool No.	Weld No.	Size(")	THK.(MM)	Weld Type	T/C No.	Remark
PQR NO. RNC-PQR-017		10"	21.44	BW	1,3	
	Total:	1		Joint.	Total:	10 DB.
	Total:			EA.	Total:	Kg.

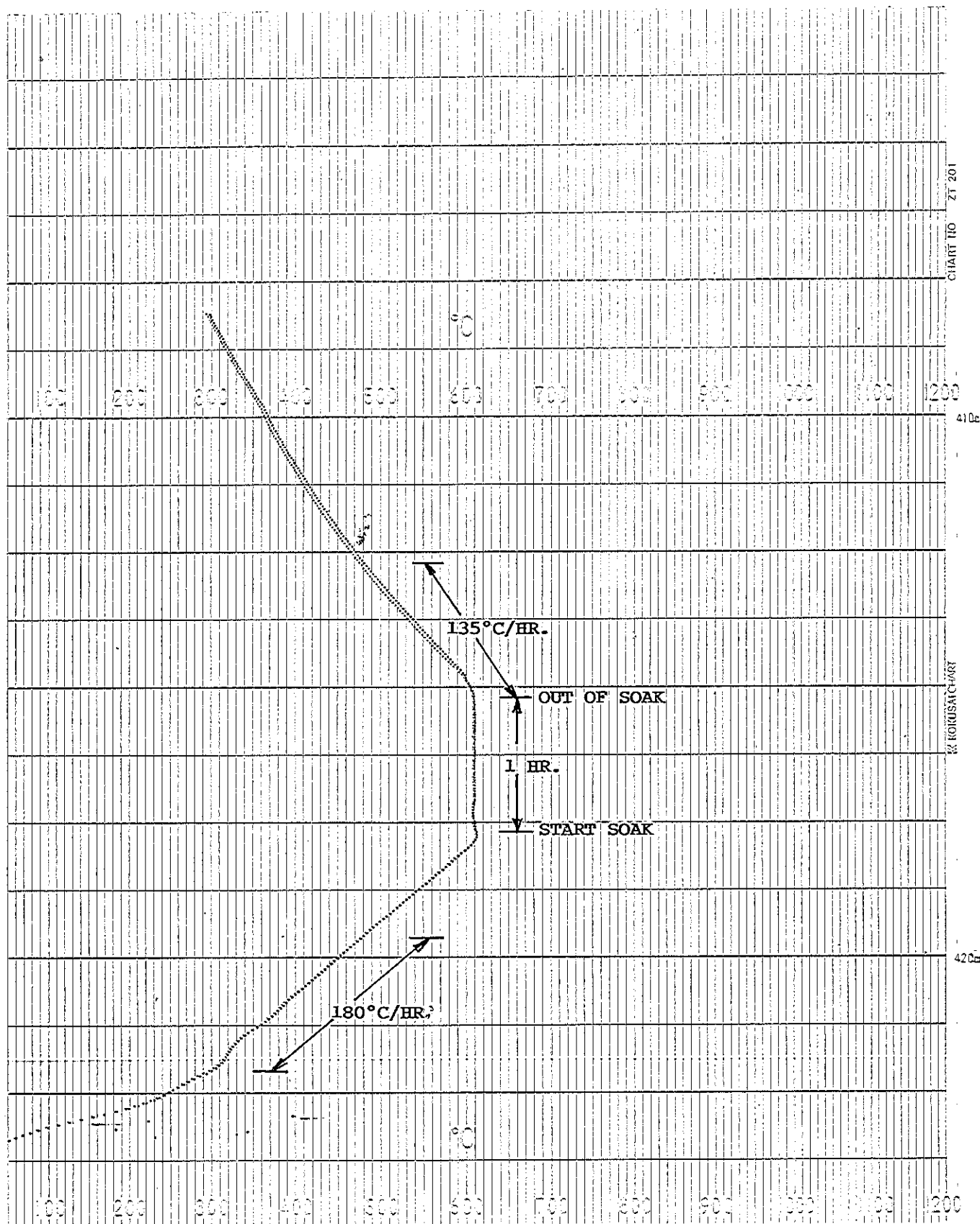
THERMOCOUPLE LOCATION



HEAT TREATMENT PROFILE



	Global Heat Treatment (Thailand) Co., Ltd.			
	Checked By	Approved By	Approved by	Approved by
Signature:				
Name:	KOMKHUN D.	Mr. Chatchanan S.	ภูชิน สีหอมไชย	ภูชิน สีหอมไชย
Date:	15-04-2013	19/04/13	Phuchin Seehomchai	Phuchin Seehomchai



DATE 15-04-2013

TC COLOUR

1. RED

3. SKY BLUE

CHART NO. PW - 003

RECORDER NO. EH 07ZA194

CHART SPEED 25mm./HR.

CUSTOMER. RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

PROJECT. MINI SHUT DOWN PTG

PQR NO. RNC-PQR-017

SIZE 10" THK. 21.44mm.

MATERIAL A106 GR.B

START TIME. 10.00 AM.

TECHNICIAN. MR. SITHIKORN P.

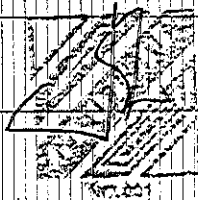
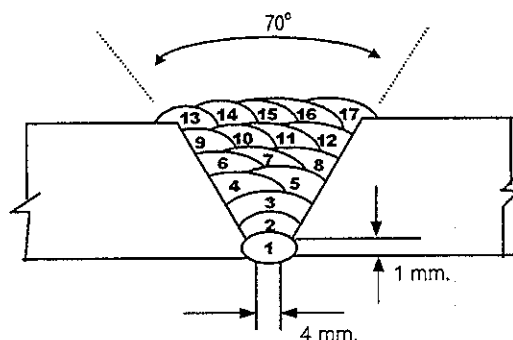


CHART NO. 71 201

PAE TECHNICAL SERVICE	WELDING PROCEDURE	PQR No. : RNC-PQR-017
		WPS No. : RNC-WPS-017
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST CONDITION AND RESULT

Welding Sequences for RNC-PQR-017 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or Weave Bead	Wire Feed Speed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
1	GTAW	ER 70S-G	2.4	100	10	DCEN	-	-	5.68	10.57	14	-
2	GTAW	ER 70S-G	2.4	125	11	DCEN	-	-	9.70	8.51	14	-
3	SMAW	E 7016	3.2	119	24	DCEP	-	-	7.96	21.53	-	-
4	SMAW	E 7016	3.2	120	24	DCEP	-	-	8.51	22.31	-	-
5	SMAW	E 7016	3.2	122	24	DCEP	-	-	8.61	20.41	-	-
6	SMAW	E 7016	3.2	121	24	DCEP	-	-	9.48	18.38	-	-
7	SMAW	E 7016	3.2	122	24	DCEP	-	-	9.62	18.27	-	-
8	SMAW	E 7016	3.2	124	24	DCEP	-	-	9.75	18.32	-	-
9	SMAW	E 7016	3.2	125	24	DCEP	-	-	9.82	19.10	-	-
10	SMAW	E 7016	3.2	120	24	DCEP	-	-	8.85	19.53	-	-
11	SMAW	E 7016	3.2	118	24	DCEP	-	-	8.58	19.81	-	-
12	SMAW	E 7016	3.2	121	24	DCEP	-	-	8.69	20.05	-	-
13	SMAW	E 7016	3.2	120	24	DCEP	-	-	9.48	18.23	-	-
14	SMAW	E 7016	3.2	123	24	DCEP	-	-	9.64	18.38	-	-
15	SMAW	E 7016	3.2	122	24	DCEP	-	-	9.36	18.77	-	-
16	SMAW	E 7016	3.2	120	24	DCEP	-	-	11.38	15.19	-	-
17	SMAW	E 7016	3.2	123	24	DCEP	-	-	9.60	18.45	-	-